

DEEPAYAN BASU

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EDUCATION

Indian Institute of Technology, Jodhpur <i>B.Tech. in Materials Engineering</i>	Jul 2021 - May 2025 <i>Jodhpur, India</i>
South Point High School (CBSE) <i>Senior Secondary / Class XII (94%)</i>	Apr 2019 - Mar 2021 <i>Kolkata, India</i>
South Point High School (CBSE) <i>Secondary / Class X (96%)</i>	Apr 2017 - Mar 2019 <i>Kolkata, India</i>

EXPERIENCE

KPMG Global Services <i>Associate Consultant - Model Risk Management (MRM), Advisory</i>	Nov 2025 - Present <i>Bengaluru, India</i>
Model Validation & Risk Diagnostics: Validated PD, LGD, EAD models (Markov Chain, Dynamic Transition Matrix, Cox Proportional Hazard, Super Panel Hazard) across retail and wholesale portfolios of clients through development, monitoring, and annual review cycles. Supported migration of legacy credit models and delivered GenAI model validation training initiatives for internal advisory teams. Model Optimization & Explainability: Applied SHAP and Gini diagnostics to evaluate feature importance and model interpretability; engineered benchmark models that outperformed production baselines by 5%+. Quantitative Diagnostics: Assessed model performance utilizing KS-statistic, Population Stability Index (PSI), Characteristic Stability Index (CSI), UPAI, and MPAI with rigorous statistical significance testing, variable bucketing, and LGD recovery analysis.	
Scientific Investing <i>LLM Intern</i>	Nov 2024 - Feb 2025 <i>Remote, India</i>
RAG & Document QA Architecture: Engineered a PDF-based financial Question-Answering system (LangChain, PyMuPDF, FinBERT, ChromaDB); achieved a 40%+ improvement in answer relevance using advanced RAG techniques with nomic-embed-text, deployed on an MCP server with 2x faster response times. Knowledge Graph Integration: Integrated Neo4j knowledge graphs for relational QA across 1,000+ financial entities, improving entity extraction and query accuracy by 30%.	
DRDO, Defence Laboratory <i>Research and Development Intern (Govt. of India)</i>	May 2024 - Jul 2024 <i>Jodhpur, India</i>
Deep Learning Architecture: Designed a dual-branch CNN (J-CNN) with task-driven pruning for SAR target classification, achieving 94% accuracy and 23% faster inference on the MSTAR dataset. Material Modeling: Modeled Radar Absorbing Materials (RAM), achieving a 30% reduction in radar cross-section (RCS) for stealth aircraft applications.	

TECHNICAL PROJECTS

Real-Time Stock Trading System Event-Driven Backend Engineering	Mar 2026 - Present
Real-Time Data Streaming: Architected a highly scalable, event-driven stock brokerage backend using FastAPI, Redis Pub/Sub, and Server-Sent Events (SSE) to stream real-time market data at 20,000+ peak QPS with sub-millisecond distribution latency. Transaction-Safe Order Processing: Engineered transaction-safe order ingestion with Apache Kafka and PostgreSQL using the Outbox Pattern for ACID compliance, leveraging TimescaleDB continuous aggregates to compress tick data by 90% for low-latency chart rendering.	
High-Throughput LLM Serving Lab (vLLM) LLM Optimization & Infrastructure	Mar 2026 - Apr 2026
- Architected an end-to-end benchmarking harness transitioning SmolLM-135M from Hugging Face inference to highly optimized vLLM topologies. - Simulated PagedAttention vs. contiguous memory allocation, recovering ~80% of KV-cache fragmentation and boosting token utilization to ~95%. - Built an OpenAI-compatible FastAPI server integrated with a live Gradio telemetry dashboard, load-tested to support 20 concurrent requests.	
Conference Call Sentiment Analysis Tool FinTech AI Pipeline	Oct 2024 - Nov 2024
Developed a VADER/Afinn sentiment analysis pipeline on 50+ earnings transcripts (PyPDF2, Pandas, NLTK), generating quarter-over-quarter sentiment vectors that achieved a 73% correlation with post-call stock movements.	
Causal ML for Travel App Participation Applied Econometrics	Sep 2024 - Dec 2024
Applied Causal Trees and Causal Forests to estimate Conditional Average Treatment Effects (CATE) across 600+ survey users, identifying a high reward-responsiveness segment (+3.36 CATE) in the 14-20 age demographic.	
Lankford Coefficient Prediction Deep Learning in Metallurgy	Jan 2024 - Mar 2024
Built a CNN to predict Lankford coefficients directly from ODF images (10,000+ segments), achieving R² scores of 0.89/0.92/0.94 and reducing processing latency by 80%.	

SKILLS

Data Science & Risk Modeling: Python, SQL, Scikit-learn, TensorFlow, Statsmodels, PyCaret, PD/LGD/EAD Frameworks
Model Diagnostics: KS-statistic, PSI, CSI, UPAI, MPAI, Gini, VIF, SHAP Explainability
GenAI & LLM Engineering: vLLM, LangChain, PagedAttention, FAISS, ChromaDB, Neo4j, FinBERT, Ollama
MLOps & Infrastructure: FastAPI, Docker, Git, GitHub Actions, MCP Server, Streamlit, Gradio, Power BI, Plotly

ACHIEVEMENTS, CERTIFICATIONS & LANGUAGES

Certifications: Claude Certified Architect Foundations (CCA-F)
Honors: 1st Place - NIFT National Cultural Fest Band Competition (2024); 3rd Place - IGNUS Band Competition, IIT Jodhpur (2025); multiple school-level athletic awards.
Languages: English, Hindi, Bengali (Fluent); Japanese (JLPT N5)

POSITIONS OF RESPONSIBILITY

Mentor & Core Team Member Sangam Music Society, IIT Jodhpur	Jul 2022 - Apr 2025
Led music production and coordinated logistics for 10+ cultural events.	
Assistant Head UDBHAAS 2022, IIT Jodhpur (JCKIF)	Nov 2022
Orchestrated volunteer operations, hospitality, and media logistics for a 500+ attendee national art exhibition.	